Emily Steiner

PhD Candidate, Stanford University | 3D/4D Computer Vision

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Research Interests

My research develops methods to uncover and learn from temporal patterns in complex, long-term observations, supporting understanding and reasoning about evolving environments. This work brings together 3D/4D scene understanding, grounded spatiotemporal representations, and vision-language models, with applications in embodied AI, robotics, and construction.

Education

Stanford University Sept 2023–2028 (expected)
MS–PhD Candidate, Electrical Engineering | Advisor: Prof. Iro Armeni, Gradient Spaces Lab.

University of Waterloo Sept 2018– June 2023
BASc, Honours Mechatronics Engineering, Computing Option, Co-op
With Distinction; Dean's Honour List. Cumulative average: 95.6%.

Publications

ReScene4D: Temporally Consistent Semantic Instance Segmentation of Evolving Indoor 3D Scenes.
Emily Steiner, Jianhao Zheng, Henry Howard-Jenkins, Chris Xie, Iro Armeni.
CVPR 2026. Paper | Project | Code

Relevant Experience

Stanford University, Gradient Spaces Lab Apr 2024–present
Graduate Research Assistant | Advisor: Prof. Iro Armeni

- Developed **ReScene4D**, a query-based transformer whose joint spatiotemporal feature encoding and query decoding improve temporal identity consistency over independent 3D segmentation with ad hoc matching.
- Introduced t-mAP to measure segmentation and temporal identity consistency; showed that sharing features across revisits also improves single-scan segmentation.

Waymo Jun–Sep 2026
Technical Intern | Mentor: Chris Ying

- Fine-tuned the Gemma4 vision-language model with LoRA in JAX for fine-grained video action recognition, achieving +15 pp mAcc over zero-shot; evaluated temporal context for long-tail scenarios.
- Designed a labelling policy and hierarchical classification task that outperformed single-label prediction while eliciting structured reasoning without chain-of-thought supervision.
- Built an end-to-end video annotation pipeline integrating multi-target tracking, temporal label propagation, and dataset curation for data-efficient training.

Stanford University, Computational Imaging Lab Jan–Apr 2024
PhD Rotation Student | Advisor: Prof. Gordon Wetzstein

- Built an audio simulator comparing head- and gaze-directed beamforming for hearing aids.

Selected Projects

Crowdsourcing 4D Data from Evolving Living Spaces Winter 2026

- Built a 3D/4D data pipeline using Swift/ARKit LiDAR scans, CLIP/Concerto validation, and Django/AWS.

Probing the Object Awareness of 3D LLMs Fall 2024

- Identified density-induced hallucination and object detection bottlenecks through 3D LLM perturbation tests.

Open Scene Understanding for Construction Analysis Spring 2024

- Lifted GroundingDINO, SAM, and CLIP semantics into 3D meshes for zero-shot construction monitoring.

Honours, Awards & Grants

TomKat Graduate Fellowship for Translational Research 2025–2027
CIFE Seed Grant | Research staff; PI: Prof. Iro Armeni 2025–2026
Queryable spatiotemporal 4D representations for construction progress monitoring.
Ansys Design Analysis Competition, 1st place, Waterloo 2023
Drag Reduction System Automation capstone with Williams Racing Formula 1.
Co-op Student of the Year, Honourable Mention, Waterloo 2022
First in Class Engineering Scholarship, Waterloo 2022, 2021, 2019

Teaching & Academic Service

3DV 2027 | Student Organizer Current
Stanford CEE 147C/247C | Teaching Assistant 2026–present

- Teaching assistant for Computer Vision for the Built Environment; leading the Winter 2027 course redesign.

Defining What Construction AI Needs to See | Lead Workshop Organizer Sept 2026

- Led 70+ industry participants in evaluating 3D job-site scans and defining scene-understanding priorities.

Computer Vision for the Built World | Workshop Organizer CVPR 2026
Nothing Stands Still | Challenge Organizer ICRA 2025; CVPR 2026
Stanford Women in Electrical Engineering 2024–2026

- Faculty Liaison (2024–2026); Co-Mentorship Chair (2025–2026); graduate mentor (2024–2026).

Additional Experience

Stanford University, Molecular Imaging Instrumentation Lab Fall 2023
PhD Rotation Student / Advisor: Prof. Craig Levin

- Studied conditional GANs for positron emission tomography (PET) image correction.

Lumafield May–Aug 2022
Hardware Research & Development Engineering Co-op

- Built a multi-detector CT prototype with iterative calibration and OpenCV fusion, doubling capture speed.
- Separately optimized the sharpness–scan-time trade-off, enabling 3× faster scans without quality loss; deployed improvements to customer machines.

Inertia Product Development Jan–Apr 2022
Product Development Intern, Mechatronics

- Built a customer-facing LiDAR point-cloud visualizer in Python/PyQt with a ROS backend.

Canadensys May–Aug 2021
Aerospace Engineering Intern

- Developed STM32 firmware and PID control for a high-precision BLDC motor; diagnosed unstable Hall-effect sensor feedback and implemented a software correction that salvaged the PCB design.

exactEarth (Spire) Jan–Apr; Sep–Dec 2020
Software Engineering Co-op

- Developed autonomous operations software and supported commissioning for EV10 (launched Sept 2020).

Technical Skills

Methods: Transformers, vision-language models, 3D point-cloud processing, multimodal learning.
Programming & ML: Python, PyTorch/Lightning, JAX/Flax, C/C++, ROS.
Training infrastructure: Distributed training (DDP/FSDP), HPC clusters, Weights & Biases, TensorBoard.

Selected Coursework

Vision & ML: Diffusion & Large Vision Models; 3D/4D Foundation Models; 3D Perception & Reconstruction; Deep Learning for Computer Vision; Virtual Reality.
Robotics: Autonomous Mobile Robots; Embedded Systems; Sensors; Automatic Control Systems.